Ideation Phase

Empathize & Discover — Empathy Map Canvas

Date	31 January 2025
Project Name	PlantCare AI
Maximum Marks	4 Marks

About PlantCare AI

PlantCare AI is an AI-powered plant disease detection system that allows farmers and gardeners to upload a leaf image and instantly receive a diagnosis (disease name, confidence %, severity, and remedy). It supports 15 disease classes across Tomato, Potato, and Pepper plants, using a TensorFlow deep-learning model served via a Flask web application.

Primary User Persona: The Smallholder Farmer / Home Gardener

THINK & FEEL

- Worried about losing crops and income
- Anxious when plant leaves look different
- Hopes for an easy and affordable solution
- Doubts whether online advice is reliable
- Wants to do right by the crop without wasting money on wrong treatments

SEE

- Neighbours losing crops to unknown diseases
- Expensive pesticides advertised in agri-shops
- Sparse internet connectivity in rural areas
- Other farmers using smartphones for basic tasks
- Market prices fluctuating based on crop yield

HEAR

- Friends suggesting traditional remedies
- Agri-extension officers giving generic advice
- News about new plant diseases spreading
- Family pressure to protect the harvest
- Retailers pushing expensive chemical solutions

SAY & DO

- Searches YouTube for leaf disease identification
- Asks neighbours to look at affected plants
- Tries home remedies based on past experience
- Delays treatment due to uncertainty about diagnosis
- Takes photos of sick plants with smartphone

PAIN

- Cannot afford expert agronomist visits
- Late diagnosis leads to total crop loss
- Confusing medical jargon in existing resources
- No offline or low-bandwidth solution available
- Fear of applying wrong pesticide and worsening situation

GAIN

- Instant, accurate disease detection from a photo
- Clear, actionable remedy instructions
- Severity rating to prioritise urgency
- Affordable — free web app, no app store needed
- Confidence to act quickly and save the crop

Key Insights

- Farmers need **instant visual feedback** — they cannot wait days for lab results.
- The remedy must be **plain-language and actionable**, not technical jargon.
- Severity classification (None / Moderate / High / Critical) helps users **prioritise urgency** and act fast.
- A **web-based interface** (no app install) lowers adoption friction for low-tech users.
- Supporting **multiple crops** (Tomato, Potato, Pepper) covers the most common smallholder plants.